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Database Management System

Basics of DBMS :-

Introduction of DBMS :-

- DBMS is a software system that manages, stores and retrieves data efficiently in a structured format. It acts as a bridge between a central database and multiple clients including apps and users.

- DBMS connects the central database with multiple clients.
- It allows user to create, update and query database efficiently.
- Ensures data integrity, consistency and security across multiple users and applications.
- Reduces data redundancy and inconsistency through centralized control.

Problems with Traditional File-Based systems :-

(i) Data Redundancy :- Duplicate entries across files.
(ii) Inconsistency :- conflicting or outdated information.
(iii) Difficult access :- Manual file search required.
(iv) Poor security :- No control over data access.
(v) No-Backup/Recovery :- Data loss was often permanent.

Components of DBMS Applications :-

1) Hardware :-
- Physical devices like servers, disks, input devices, network devices etc
- stores and processes data, interfaces between real-world inputs and digital systems.

2) Software :-
- Actual DBMS softwares like MySQL, oracle, PostgreSQL.
- Includes the database engines, OS, network software & application tools
- Translates, database access language into operations.

3) Data :-
- Raw facts stored in structured or unstructured formats.
- Operational data :- Actual user data (eg, name, age)
- Metadata :- Data about data. (eg, storage time, size, data type)

4) Procedures :-
- Instructions and rules for using DBMS efficiently
- Covers setup, login/logout, data validation, backup, access control, & report generation.
- Helps to ensure secure & consistent use of systems.

5) Database Access Language :-
- Used to interact with the database (Create, read, update, delete)
- Ex :- SQL, Myaccess, oracle PL/SQL.
- Data Definition Language :- CREATE, ALTER, DROP
- DML :- INSERT, UPDATE, DELETE

6) People :-
- Users interacting with DBMS at different levels.
- Database Administrator :- Manage security, performance, user access.
- Developers :- Build applications using the databases.
- End Users :- User applications to access the database.

Types of DBMS

1. RDBMS
2. NoSQL DBMS
3. OODBMS
4. Hierarchical Database
5. Network based Database
6. Cloud-based database

1) Relational Database Management System :-
- It organizes data into tables composed of rows & columns.
- Uses primary keys to uniquely identify rows and foreign keys to establish relationship between tables.
- Queries are written in SQL, which allows for efficient data manipulation & retrieval.
- Ex :- MySQL, Oracle, Microsoft SQL Server and PostgreSQL.

2) NoSQL DBMS :-
- They are designed to handle large scale data and provide high performance for scenarios where relational models might be restrictive.
- They store data in various non-relational formats, such as key-value pairs, documents, graphs or columns.
- These flexible data models enable rapid scaling and are well suited for unstructured or semi structured data.
Ex :- MongoDB, Cassandra, DynamoDB & Redis.

3) Object-oriented DBMS (OODBMS) :-
- It integrates object oriented programming concept into database environment, allowing data to be stored as objects.
- Supports complex data types and relationships, making it ideal for applications requiring advanced data modeling and real-world simulations.
- Ex :- objectDB, db4o.

4) Hierarchical Database :-
- organizes data in tree like structure, where each record has a single parent and have multiple children.
- This model is similar to a file system with folders & subfolders.
- It is efficient for storing data with a clear hierarchy, such as organizational charts or file directories.
- Navigation is fast and predictable due to the fixed structure.

- It lacks flexibility & is difficult to restructure or handle complex many to many relationships.
- Ex:- IBM Information Management system.

5) Network Database :-
- It uses a graph like model to allow more complex relationships between entities.
- Unlike the hierarchical model, It permits each child to have multiple parents, enabling many-to-many relationships.
- Data is represented using records & sets, where sets define the relationships.
- Ex:- Integrated Data store, TurboIMAGE.

6) Cloud-Based Database :-
- They are hosted on cloud computing platforms like AWS, Azure or Google Cloud.
- They offer on demand scalability, high availability, automatic backup & remote accessibility.
- Ex:- Amazon RDS, MongoDB, Google BigQuery

* Database Languages :-

Database languages are specialized sets of commands & instructions used to define, manipulate and control data within a database. Each language type plays a distinct role in database management, ensuring efficient storage, retrieval and security of data.

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Types of DBMS Languages

Data Definition Language	Data Manipulation Language	Data control Language	Transaction control Language	Data Query Language
- CREATE	- INSERT	- REVOKE	- ROLLBACK	- SELECT
- DROP	- DELETE	- GRANT	- COMMIT	
- TRUNCATE	- UPDATE			
- RENAME				
- ALTER				

1) Data Definition Language :-

- DDL is the short name, which deals with database schemas and descriptions, of how the data should reside in the database.
- CREATE :- to create a database and its objects like (table, index, views)
- ALTER :- alters the structure of existing database.
- DROP :- delete objects from the database.
- TRUNCATE :- remove all records from a table, included space allocated.
- COMMENT :- add comments to the data dictionary.
- RENAME :- rename an object.

2) Data Manipulation Language :-

- It focuses on manipulating the data stored in database, enabling users to retrieve, add, update and delete data.
- INSERT :- insert data into a table
- Update :- updates existing data within a table
- DELETE :- delete all records from a database table
- Merge :- upsert operation
- Call :- call a PL/SQL or Java subprogram
- Lock Table :- concurrency control.

3) Data Control Language :-

- It manages access permisions, ensuring data security by controlling who can perform certain actions on database.
- GRANT :- Provides specific privileges to user (eg, SELECT, INSERT)
- REVOKE :- Removes previously granted permissions from a user

4) Transaction Control Language :-

- TCL commands oversee transactional data to maintain consistency reliability and atomicity
- ROLLBACK :- Undoes changes made during a transaction.
- COMMIT :- Saves all changes made during a transaction.
- SAVEPOINT :- Sets a point whithin a transaction to which one can later roll back

5) Data Query Language :-

- DQL is a subset of SQL used to retrieve data from a database without modifying it
- It's main command is SELECT., which allows users to fetch specific information based on their requirments.

Applications Of DBMS :-

- Banking :- Manages accounts and transactions
- E-commerce :- Tracks Products, orders and customers.
- Health-care :- Stores patient records and diagnosis.